Beyond the Averages: A Multivariate Analysis of Identity, Environment, and Mental Health in College Students

Final Project

Brandon Tran (M.S. S&DS)
brandon.tran@yale.edu

April 30, 2026

1 Introduction

Over the course of the semester, we studied the Healthy Minds Networks' latest 2024–2025 Healthy Minds Study (HMS) student survey. The HMS is a population-level, web-based survey administered by the HMN to assess mental health, service utilization, and related factors among post-secondary students. Since its inception over 15 years ago, the study has collected more than $935,000$ responses from over 675 institution nationwide. The latest 2024–2025 iteration was graciously made available to us by the University of Michigan School of Public Health, containing $n = 84,735$ individual respondents across $1,608$ questions.

This study examines the multivariate relationship between marginalized identities, environmental exposures, and mental health outcomes in the modern college student population. Utilizing the HMS survey's standard Demographics and Mental Health Status modules, we move beyond isolated univariate metrics to map the complex topology of student distress. Specifically, this analysis investigates the latent linear structure of psychological burden, identifies the specific environmental catalysts that drive severe non-linear symptoms, and quantifies the exact mental health disparities experienced by queer and gender-expansive students. Finally, by evaluating whether distress profiles can accurately predict a student's identity group, we reveal the clinical overlap shared across the broader student body.

2 Design and Primary Questions

The primary objective of our research is to move beyond isolated metrics of mental health to understand the complex, multivariate reality of student well-being. Specifically, we aim to answer three core questions:

1. How do specific clinical symptoms of anxiety and depression structurally co-occur, and what environmental stressors drive their most severe manifestations?
2. To what extent do intersecting identities (specifically, sexual orientation and gender identity) and institutional interventions (specifically, therapy) alter a student's multivariate mental health profile?
3. Are these psychological differences distinct enough that a student's mental health profile can accurately predict their marginalized identity group?

To answer these questions and uncover the relationships within the data, we employ three multivariate techniques. Each method builds upon the findings of the last:

- **Ordination (CA & NMDS):** We will use ordination to non-linearly map the topological structure of specific, individual mental health symptoms. By overlaying environmental stressors (like financial burden or internet usage) onto this map, we can pinpoint exactly which external factors drive the most severe symptom clusters. This establishes our baseline understanding of how distress manifests.
- **MANOVA:** Having mapped the structural and environmental realities of student distress, we will pivot to analyzing group differences. We will use MANOVA to test whether specific demographics (sexual orientation) and interventions (therapy utilization) significantly alter the mean vectors of these mental health outcomes simultaneously.
- **Discriminant Analysis:** If MANOVA confirms that mental health outcomes differ significantly across demographic groups, we will reverse the lens. Using Discriminant Analysis, we will test whether a student's psychological profile is distinct enough to accurately predict their gender and sexual identity group. Misclassifications here will be just as insightful as accurate predictions, as they will help us identify shared systemic stressors and psychological overlap between marginalized groups.

Note: We use a *fourth* multivariate method, Principal Component Analysis (PCA) as part of upfront Section 4, to extract the primary linear dimensions from our transformed composite scores, establishing a baseline structural understanding of how anxiety, depression, and flourishing share variance. This will serve as a foundational understanding to answer the research questions we have noted above.

3 Data and Variables

As mentioned in Section 1, the dataset used for this analysis originates from the 2024–2025 Healthy Minds Study (HMS). The HMS is a population-level, web-based survey administered to post-secondary students to assess mental health, service utilization, and related lifestyle factors. For the purpose of this analysis, we focus on $n = 84,735$ individual respondents across questions that fall under the Standard Modules.

3.1 Overview of Variables

Table 1 summarizes the composite continuous variables, individual symptom frequencies, and categorical predictors used in our models:

Table 1: Description of HMS Variables

Variable	Description	Type	Range
Composite Scores			
flourish	Flourishing and well-being	Continuous	8 - 56
deprawsc	Depression severity (PHQ-9)	Continuous	0 - 27
anx_score	Anxiety severity (GAD-7)	Continuous	0 - 21
lonesc	Loneliness severity (UCLA)	Continuous	3 - 9
ed_sde	Disordered eating behaviors	Continuous	0 - 5
Individual Symptoms			
phq9_1 to phq9_9	Individual depression symptoms	Ordinal	1 - 4
gad7_1 to gad7_7	Individual anxiety symptoms	Ordinal	1 - 4
Environmental / Lifestyle			
exerc	Average weekly exercise	Ordinal	1 - 5
internet_1	Time spent online	Ordinal	1 - 8
fincur	Financial stress level	Ordinal	1 - 5
sleep_wknight	Average weeknight sleep	Continuous	Numeric Hours
Identity / Predictors			
queer	Sexual orientation	Categorical	0 (Cis-Het) / 1 (Queer)
ther_any	Therapy utilization	Categorical	0 (No) / 1 (Yes)
identity_group	Gender/sexual minority identity	Categorical	Cis-Het, Cis-LGBQ+, TGNC

3.2 Sources of Error and Questionable Points

Because this data is entirely self-reported, it is inherently subject to recall bias (e.g., estimating exact hours of sleep or recalling feelings ”over the last 2 weeks”). Furthermore, as we will explore in our assumption checks, the bounded nature of Likert-scale survey questions creates severe floor and ceiling effects. Many of the composite variables (such as depression and anxiety) contain a high density of ”0” responses, leading to distinct violations of multivariate normality that require careful handling or robust, non-parametric alternatives.

4 Descriptive Plots and Summary Statistics

Before proceeding with our multivariate analyses, we must understand the underlying distribution and relationships between our composite variables: flourishing (`flourish`), depression (`deprawsc`), anxiety (`anx_score`), loneliness (`lonesc`), and disordered eating (`ed_sde`).

4.1 Univariate Distributions

First, we analyze the univariate distributions of our variables to identify any skewness or necessary transformations.

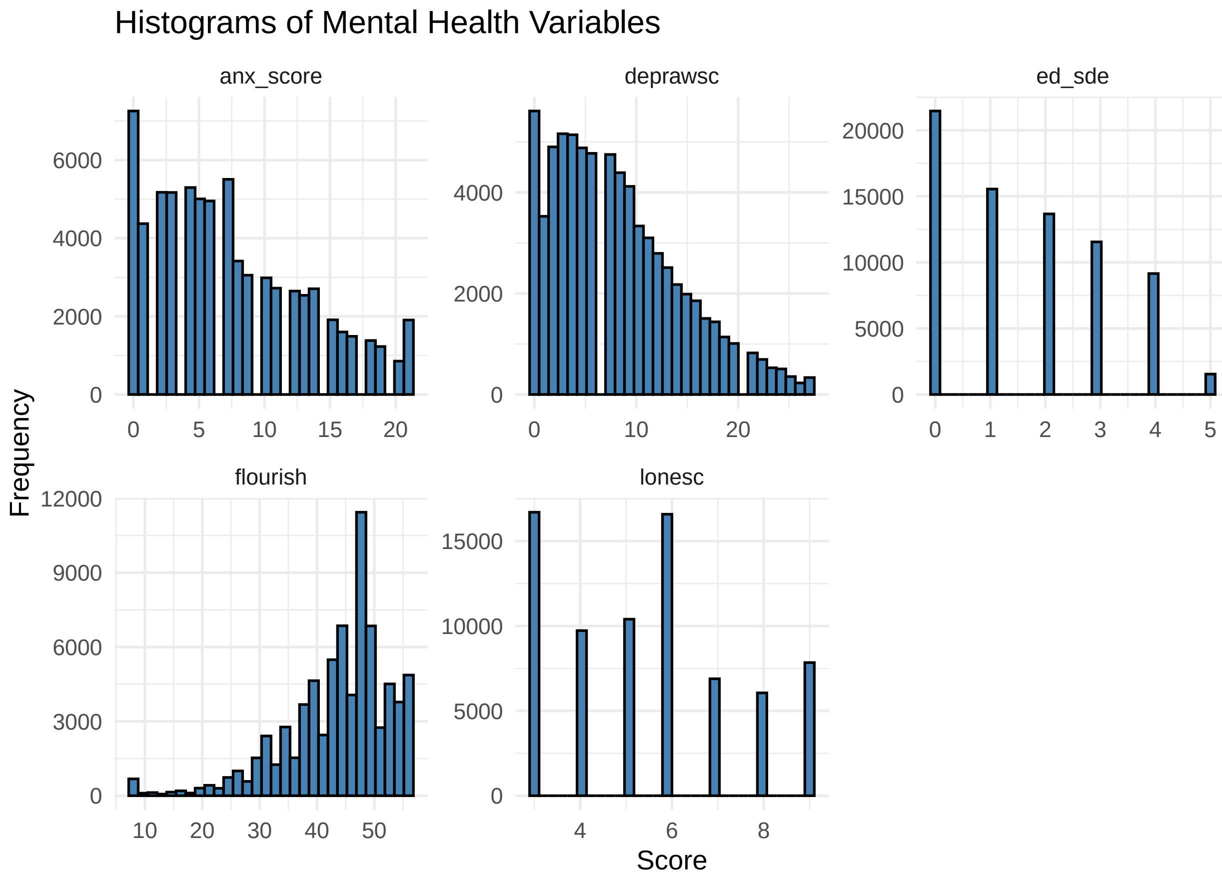

Figure 1: Histograms reveal skewness in composite variables.

The histograms in Figure 1 reveal distinct skewness in our data. Anxiety, depression, and disordered eating all exhibit heavy right skews, driven by the floor effects of the Likert-scale questions (a result of many students reporting zero or very few symptoms). Conversely, flourishing exhibits a left skew. Loneliness retains a lightly multimodal structure due to its smaller subset of underlying questions.

4.2 High-Dimensional Linearity

Despite the skewed distributions, our initial correlation plot suggests (Figure 2) strong linear relationships between our composite variables.

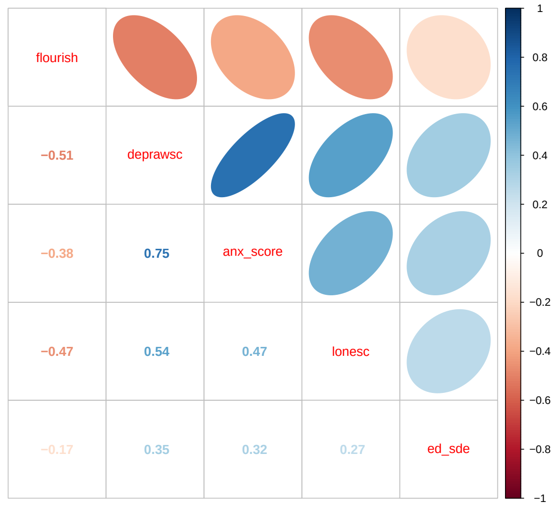

Figure 2: Correlation plot suggests strong linear relationships between composite variables.

Observe a distinct cluster of strong positive correlations between depression and anxiety ($r = 0.75$), as well as loneliness. These negative psychological indicators are all strongly negatively correlated with flourishing (e.g., flourishing and depression at $r = -0.51$). To visually confirm this high-dimensional linearity, we utilize the `PerformanceAnalytics` package to generate a matrix plot. We sample $5,000$ observations for this plot, as plotting all $68,062$ observations would result in severe overplotting and be computationally intensive.

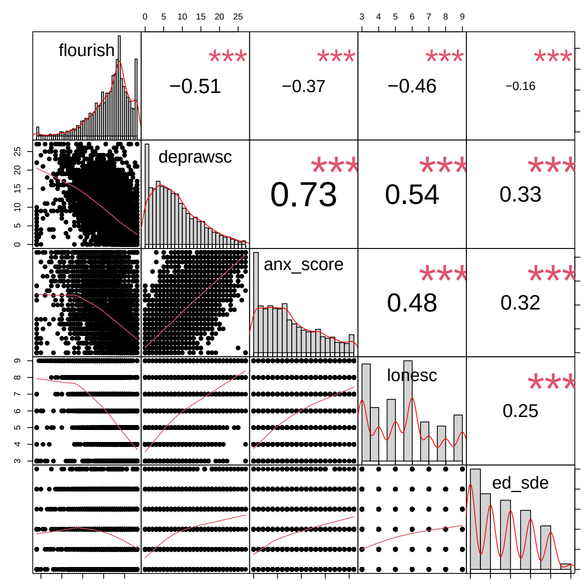

Figure 3: High-dimensional linearity is confirmed via a matrix plot.

The resulting matrix plot in Figure 3 remain consistent with our earlier analyses. The LOESS curves reveal that while the relationship between `deprawsc` and `anx_score` is highly linear, the relationships with `flourish` display slight non-linearity.

4.3 Multivariate Normality and Transformations

Finally, we test for multivariate normality using a Chi-Square Q-Q plot.

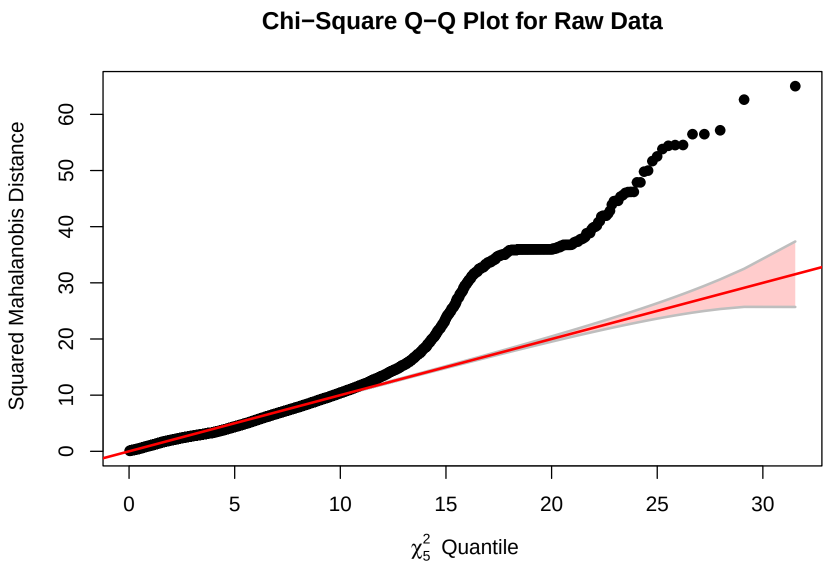

Figure 4: A Chi-Square Q-Q plot reveals the data are not multivariate normal.

We see strong structure and significant deviation from the diagonal in the upper tails, confirming that the data are not multivariate normal. At the cost of interpretability, we transform the variables, both to correct the univariate skewness observed in our histograms, as well as to improve linearity and reduce outlier influence for our subsequent methods (like PCA).

- **Square Root:** `deprawsc` and `anx_score` to correct for right-skew (incremented by 0.5 to stabilize variance for zeroes.
- **Power Transformation (x^2):** `flourish` to correct for left-skew.

To evaluate the efficacy of these transformations, we review the Chi-Square Q-Q plot for our newly transformed data.

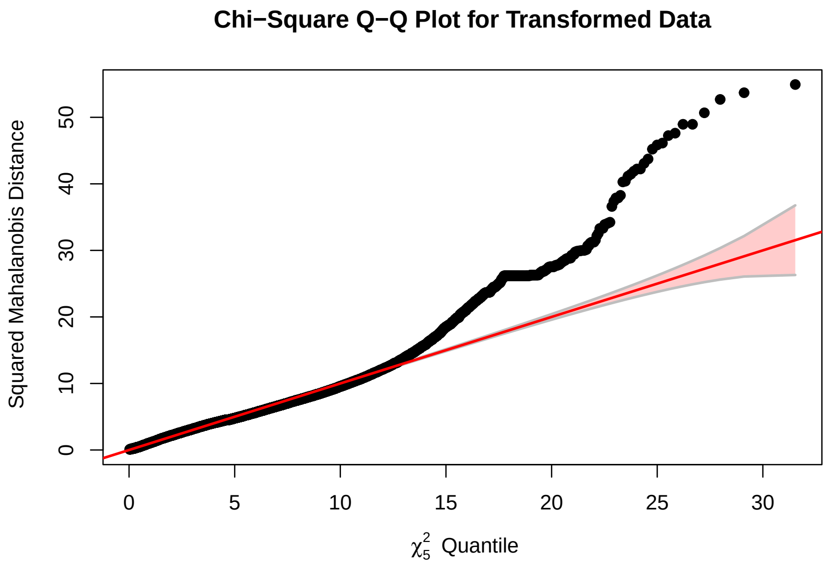

Figure 5: Enter Caption

While our data still exhibit some staircase effects inherent to bounded Likert-survey data, the heavy tail is noticeably more controlled. Furthermore, Table 2 illustrates that these transformations slightly enhanced the linear correlations across our variable pairs. By controlling these outliers and stabilizing the variance, we establish a more robust, linearized foundation for exploring group disparities and predictive classification in our subsequent multivariate models.

Table 2: Correlation Comparison (Pre vs. Post Transformation)

Variable Pair	**Original**	**Transformed**	**Change**
deprawsc_sqrt vs flourish	−0.508	−0.534	−0.025
anx_sqrt vs flourish	−0.382	−0.416	−0.035
lonesc vs flourish	−0.467	−0.504	−0.036
ed_sde vs flourish	−0.172	−0.189	−0.016
anx_sqrt vs deprawsc	0.748	0.759	0.012
lonesc vs deprawsc	0.540	0.550	0.010
ed_sde vs deprawsc	0.348	0.365	0.018
lonesc vs anx_score	0.475	0.487	0.013
ed_sde vs anx_score	0.323	0.340	0.018
ed_sde vs lonesc	0.265	0.265	0.000

4.4 Dimensionality Check: Principal Component Analysis (PCA)

By controlling the extreme outliers and stabilizing the variance through our square-root and power transformations, we establish a robust, linearized foundation optimized for Principal Component Analysis (PCA). Because PCA fundamentally relies on maximizing shared linear variance to extract latent dimensions, utilizing these transformed variables is ideal to increase the accuracy of our eigenvectors.

We conduct a PCA with scaling on the five core mental health composite scores (`flourish_sq`, `deprawsc_sqrt`, `anx_sqrt`, `lonesc`, and `ed_sde`) to evaluate the underlying dimensionality of student distress.

Table 3: Principal Component Analysis: Variance Explained

Component	Eigenvalue	% of Variance	Cumulative Variance
PC1	2.825	56.5%	56.5%
PC2	0.852	17.0%	73.5%
PC3	0.609	12.2%	85.7%

As detailed in Table 3, the first principal component (PC1) accounts for a massive 56.5% of the total variance, with an eigenvalue of 2.825 (well above the Kaiser criterion of 1.0). Reviewing the component loadings, PC1 is heavily loaded by depression (-0.527), anxiety (-0.494), and loneliness (-0.452) in the negative direction, and flourishing (0.423) in the positive direction. This confirms our hypothesis that these specific modules do not operate independently; rather, they form a single, dominant latent construct of "general psychological distress" (moving left along the axis) versus "well-being" (moving right).

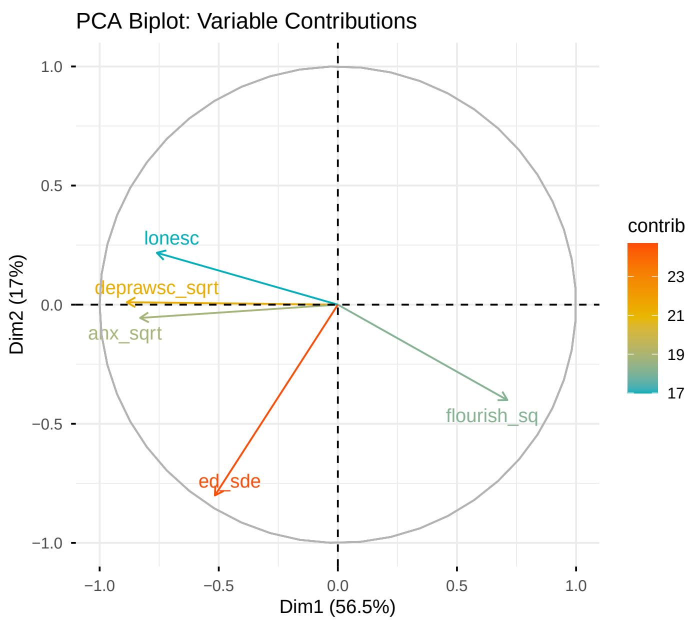

Figure 6: PCA Biplot of Transformed Mental Health Variables.

The biplot in Figure 6 visually reinforces this structure. The vectors for depression, anxiety, and loneliness travel tightly together along the primary axis, directly opposing flourishing. Interestingly, the eating disorder screener (`ed_sde`) separates heavily along the second principal component (PC2 loading of -0.867), indicating that while it is correlated with general distress, it represents a distinct behavioral dimension separate from core mood disorders.

4.5 Discussion on Transformations: Preserving Interpretability

While our PCA successfully utilized the transformed variables to prove the latent linear structure of our data, these mathematical transformations fundamentally alter the scale of the responses. A 1-point increase in a raw PHQ-9 score has direct, real-world clinical meaning; a 1-point increase in the square root of a PHQ-9 score does not.

For our primary multivariate methods (Ordination, MANOVA, and Discriminant Analysis), we intentionally revert to the raw, untransformed variables. Ordination relies on exact ordinal counts and rank-order distances that mathematical transformations would distort. Furthermore, because our massive sample size ($n > 30,000$) guarantees parametric robustness for our MANOVA and LDA models under the Central Limit Theorem, the slight statistical benefits of transforming the data do not outweigh the loss of direct interpretability. Therefore, we preserve the original, validated scoring metrics for all subsequent environmental mapping, group comparisons, and predictive classifications in Section 5.

5 Multivariate Analysis

5.1 Method 1: Ordination

We begin by performing indirect ordination to explore the underlying topological structure of mental health symptoms among students. Our goal is to reduce the dimensionality of our 16 symptom variables across the PHQ-9 and GAD-7 modules to visualize how they group together.

5.1.1 Correspondence Analysis (CA) and Inertia

We first complete a Correspondence Analysis (CA), which is ideal for our data because the symptom data represents frequency counts (scaling from 1 = "Not at All" to 4 = "Nearly Every Day"). CA maps both students and symptoms into low-dimensional space via chi-square distances.

Table 4: Correspondence Analysis (CA) Variance Explained

Dimension	Proportion of Variance	Cumulative Variance
Axis 1 (CA1)	16.48%	16.48%
Axis 2 (CA2)	10.56%	27.04%
Total Inertia	**0.1073**	

As detailed in Table 4, our total inertia is 0.1073, representing the total variance in our data (calculated as a scaled chi-square statistic). Axis 1 explains 16.48% of the variance, while Axis 2 explains 10.56%, capturing 27.04% of the variance in total.

Upon visualizing the first two dimensions of the CA (Figure 7), there is no clear evidence of an arch effect (data snaking). Rather than forming a classic U or V crescent shape, the data forms a dense, continuous cloud centered around the origin. The lack of snaking and the tight clustering of symptom variables indicate that students do not experience a complete turnover of symptoms. Instead, mental health symptoms measured by PHQ-9 and GAD-7 are highly comorbid; students scale along a dominant gradient of overall psychological distress, experiencing these symptoms simultaneously rather than in isolation.

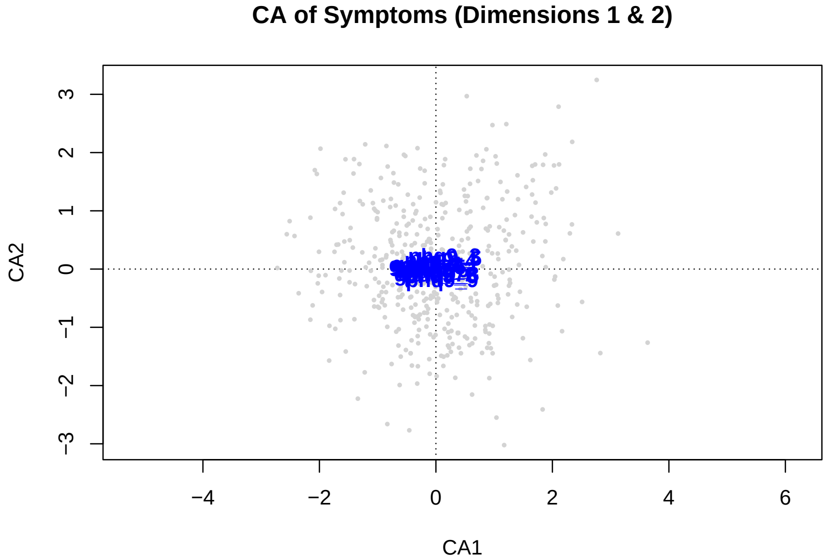

Figure 7: Correspondence Analysis of PHQ-9 and GAD-7 Symptoms.

5.1.2 Non-Metric Multidimensional Scaling (NMDS) and Stress

Because our symptom scales rely on ordinal frequency with high rates of joint absences, we pivot to Non-Metric Multidimensional Scaling (NMDS). We use a Bray-Curtis distance metric to better preserve the rank-order topology and ensure shared absences (e.g., two students both reporting "not at all" for self-harm) do not artificially inflate similarity.

To determine the appropriate number of dimensions, we evaluate the model's stress across 1D, 2D, and 3D solutions.

Table 5: NMDS Stress by Dimensionality

Dimensions (k)	**Stress Value**
1D	0.2000
2D	0.1347
3D	0.1057

As shown in Table 5, moving from a 1D to 2D solution provides a steep reduction in stress, falling well below the acceptable threshold of 0.20. While the 3D solution reduces the stress further, the 2D solution is optimal as it perfectly balances goodness-of-fit with visual interpretability.

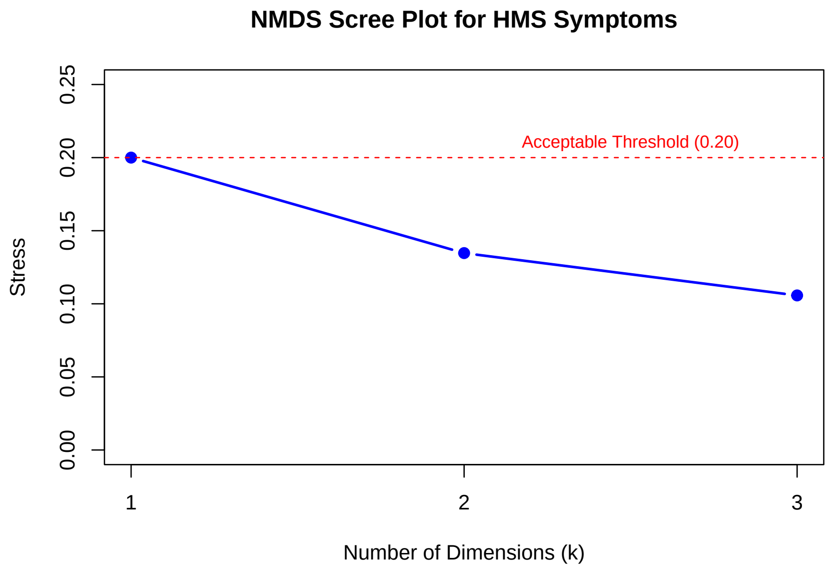

Figure 8: NMDS Scree Plot evaluating stress by dimensionality.

The 2-dimensional plot in Figure 9 reveals a clear mapping of the symptom space. Observe the extreme observation `phq9_9` on the far left-hand side, which specifically measures suicidal ideation and self-harm. The remaining general anxiety and depression symptoms form a looser cluster on the right. A notable distinction exists along the vertical axis: physical manifestations of depression like `phq9_3` (sleep) and `phq9_4` (fatigue) cluster at a lower elevation, while items situated higher on the y-axis are predominantly GAD-7 anxiety variables, such as uncontrollable worry. The NMDS algorithm successfully captured the latent structural divide between physical exhaustion, depressive symptoms, and anxious states.

5.1.3 Overlaying Environmental Variables

To understand what drives this topology, we overlay five continuous environmental variables onto our 2D NMDS solution in Figure 9: age, loneliness score (`lonesc`), exercise hours (`exerc`), internet use (`internet_1`), and financial stress (`fincur`).

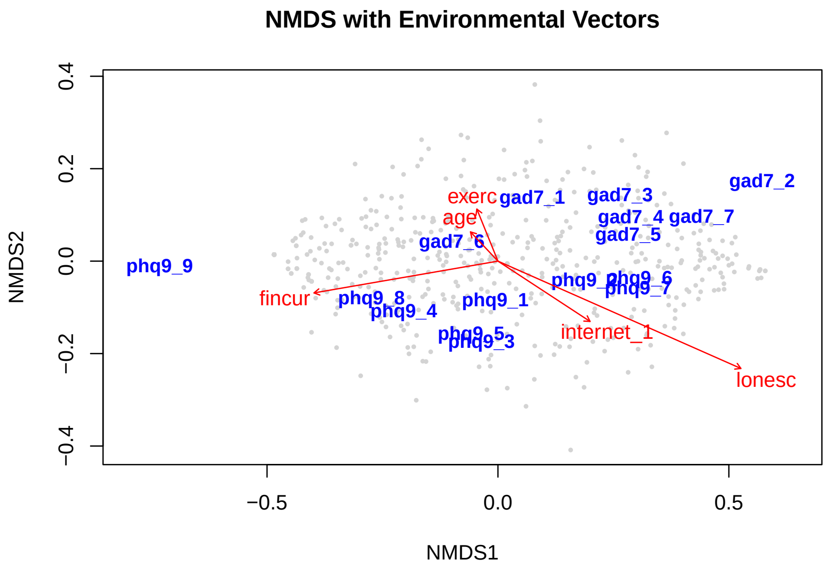

Figure 9: NMDS biplot with overlaid environmental vectors.

Through environmental vector fitting, we found that four of our five variables are statistically significant. Loneliness has an extremely high R^2 value of 0.3228, followed in magnitude by `fincur` and `internet_1`. Looking at the direction of each arrow, `lonesc` and `internet_1` point down and to the right, pulling toward the dense blend of generalized anxiety and depression symptoms. Meanwhile, `fincur` (financial stress) pulls hard to the left along the primary axis, directly toward the severe outlier `phq9_9`. This indicates that while loneliness drives general distress, acute financial burden is a major driver behind more severe suicidal ideation.

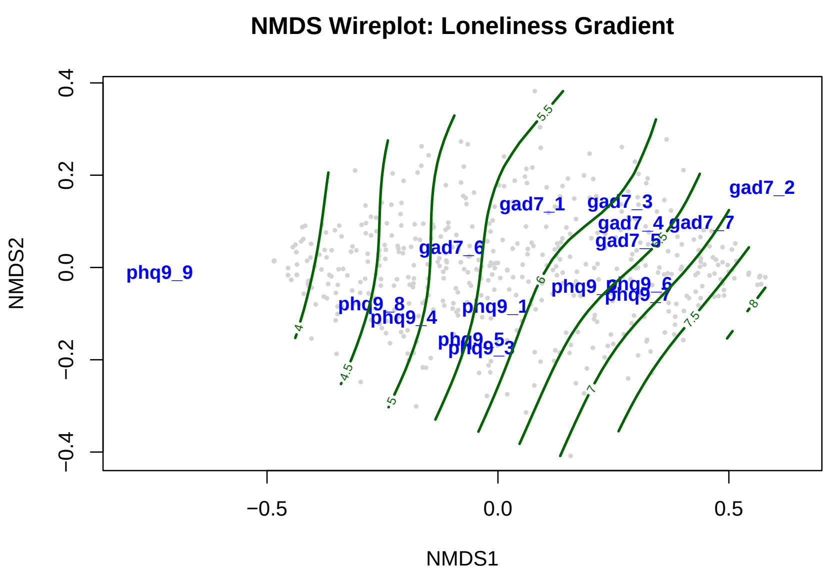

Figure 10: Non-linear GAM surface (wireplot) of loneliness mapped onto NMDS symptom space.

We further visualize this by generating a 2-dimensional Generalized Additive Model (GAM) wireplot (Figure 10). The resulting surface contours confirm that extreme loneliness scores strictly map into the dense cluster of comorbid symptoms on the right, proving that demographic and lifestyle factors are deeply entangled with the complex, non-linear architecture of a student's mental health.

5.2 Method 2: Two-Way MANOVA

With the topology of student mental health and environmental drivers mapped, we investigate how demographic and lifestyle factors independently and interactively influence these outcomes. We run a two-way Type III Multivariate Analysis of Variance (MANOVA) to examine the effect of therapy utilization (`ther_any`) and sexual orientation (`queer`) on depression and flourishing simultaneously. A Type III (Marginal) test is used as it is the most conservative approach for survey data with unequal group sizes.

5.2.1 Multivariate and Univariate Main Effects

Using Pillai's Trace, the overall multivariate model yielded statistically significant results ($p < 0.001$). While the interaction between therapy and sexual orientation was technically significant, review of the F-statistics reveals that the main effects are substantially larger than the interaction (Pillai approximate $F = 17$).

Table 6: Univariate ANOVA Results for Main Effects

Factor	Response Variable	F-Statistic	p-value
Sexual Orientation (`queer`)	Depression (`deprawsc`)	1351	< 0.001
	Flourishing (`flourish`)	734	< 0.001
Therapy Utilization (`ther_any`)	Depression (`deprawsc`)	1101	< 0.001
	Flourishing (`flourish`)	276	< 0.001

Table 7: Estimated Marginal Means: Pairwise Contrasts

Contrast	Response Variable	Estimate (Δ)	p-value
Queer $-$ Heterosexual	Depression (`deprawsc`)	$+2.71$	< 0.001
	Flourishing (`flourish`)	-2.99	< 0.001
Therapy $-$ No Therapy	Depression (`deprawsc`)	$+1.61$	< 0.001
	Flourishing (`flourish`)	-1.00	< 0.001

As detailed in Table 6 and quantified by the exact marginal contrasts in Table 7, the model is largely additive. Queer students reported significantly poorer outcomes: compared to their heterosexual peers, depression scores were on average 2.71 points higher (worse), and flourishing scores were 2.99 points lower. Similarly, students actively in therapy reported depression scores 1.61 points higher and flourishing scores 1.00 points lower compared to non-users. This implies that while therapy is associated with higher distress (likely reflecting the population seeking care), it does not fundamentally alter or close the mental health gap observed between queer students and their heterosexual peers.

5.2.2 Covariate Integration: The Role of Sleep

To contextualize these disparities against lifestyle factors, we add average hours of sleep per weeknight (`sleep_wknight`) as a continuous covariate to our model.

Somewhat surprisingly, sleep is not only a significant predictor in our multivariate analysis, but it is also the largest overall effect (Pillai approximate $F = 1383$). This indicates that healthy lifestyle factors can meaningfully offset distress; however, the mental health disparities for queer students and those in therapy are not fully explained by sleep differences alone, as those categorical predictors retained their large magnitudes of significance.

5.2.3 Evaluating Assumptions and Non-Parametric Validation

To evaluate the assumption of multivariate normality, we reviewed the squared Mahalanobis distances of our residuals.

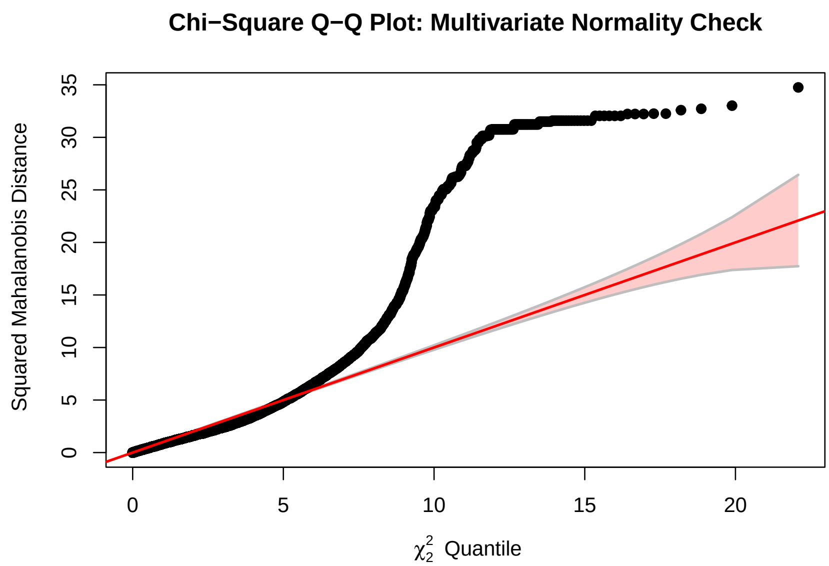

Figure 11: Chi-Square Q-Q Plot indicating violation of multivariate normality.

As seen in Figure 11, the data deviate from the theoretical χ^2 distribution in the upper tails. This is an expected artifact of the floor effects inherent to bounded Likert-scale responses. While we explored Box-Cox transformations during EDA to stabilize the variance, the sheer sample size of our data provides robustness for our parametric tests under the Central Limit Theorem.

Table 8: MRPP Non-Parametric Validation

Grouping Variable	**Agreement Statistic (A)**	**p-value**
Sexual Orientation (`queer`)	0.025	0.001
Therapy Utilization (`ther_any`)	0.008	0.001

To be rigorously thorough, we validated our findings using Multi-Response Permutation Procedures (MRPP), the non-parametric alternative to MANOVA. Using a Euclidean distance matrix on a sampled subset ($n = 2,000$), the MRPP results in Table 8 confirm significant differences across both sexual orientation and therapy utilization groups. This confirms that the observed disparities in student mental health persist across both parametric and non-parametric evaluations, regardless of distributional assumptions.

5.3 Method 3: Discriminant Analysis

Having established that marginalized identities and therapeutic interventions significantly alter mental health outcomes, we now reverse our analytical lens. We use Discriminant Analysis to determine if these complex psychological profiles are distinct enough to accurately classify a student's gender and sexual identity group (`identity_group`: Cis-Het, Cis-LGBQ+, or TGNC).

Even though we explored transformations during our exploratory data analysis, we chose to proceed with the raw, untransformed variables (depression, anxiety, flourishing, loneliness, and eating disorder symptoms) for this section to preserve the direct clinical interpretability of our classification boundaries and coefficients.

5.3.1 Assumptions and Model Selection

We first evaluated the assumption of equal covariance matrices across our three identity groups using Box's M-test. The test yielded a significant result ($p < 0.001$), indicating a violation of this assumption. Because Quadratic Discriminant Analysis (QDA) does not assume equal covariance matrices, we ran both Linear Discriminant Analysis (LDA) and QDA using leave-one-out cross-validation to compare their predictive efficacy.

Table 9: Cross-Validated Classification Accuracy

Model	**Overall Accuracy**
Linear Discriminant Analysis (LDA)	0.6723
Quadratic Discriminant Analysis (QDA)	0.6636

As shown in Table 9, both models performed similarly. Because LDA is generally robust to violations of homogeneity when sample sizes are large, and because LDA provides readily interpretable standardized coefficients (which QDA does not), we elected to proceed with the Linear Discriminant Analysis model.

5.3.2 Stepwise Selection and Discriminant Directions

To optimize our model, we performed a stepwise selection process using 10-fold cross-validation (`direction = "both"`). With a strict stopping criterion requiring at least a 5% improvement in classification rate, the algorithm only retained loneliness (`lonesc`), achieving a 67.6% correctness rate. This makes sense given our highly imbalanced prior probabilities (where guessing the majority group yields a 67% accuracy baseline). However, to understand the relative clinical contributions of the entire symptom landscape, we evaluate the standardized coefficients of the full LDA model.

Table 10 outlines the standardized coefficients for our two linear discriminants. LD1 captures the vast majority of the variance (94.8%). By evaluating the magnitude and direction of the coefficients, we see that LD1 is heavily driven by elevated depression (0.486) and loneliness (0.317) opposing flourishing (-0.268), creating a primary axis of overall psychological distress. LD2, while

Table 10: Standardized Canonical Discriminant Function Coefficients

Predictor Variable	**LD1**	**LD2**
Depression (`deprawsc`)	0.486	0.560
Anxiety (`anx_score`)	0.227	−0.574
Flourishing (`flourish`)	−0.268	−0.209
Loneliness (`lonesc`)	0.317	0.105
Eating Disorder (`ed_sde`)	−0.053	−0.933
Proportion of Trace	**94.8%**	**5.2%**

capturing much less variance (5.2%), is defined largely by eating disorder symptoms (−0.933) and anxiety (−0.574) separating from depression, offering a secondary dimension of specific clinical manifestation.

5.3.3 Classification Accuracy and Confusion Matrix

While the overall cross-validated accuracy of the LDA model provides a baseline of 67.2%, a detailed review of the confusion matrix reveals the nuanced realities of predicting identity from psychological distress.

Table 11: LDA Cross-Validated Confusion Matrix and Class Recall

	Predicted Class			
True Class	**Cis-Het**	**Cis-LGBQ+**	**TGNC**	**Recall (Sensitivity)**
Cis-Het	42,655	874	204	0.975
Cis-LGBQ+	15,068	724	130	0.046
TGNC	4,515	401	102	0.020

As detailed in Table 11, the model is highly sensitive to the `Cis-Het` group (97.5% recall) but struggles significantly to accurately classify the `Cis-LGBQ+` (4.6%) and `TGNC` (2.0%) populations. The vast majority of misclassifications for queer and trans students are predicted as Cis-Het. This is largely an artifact of the prior probabilities in our highly imbalanced dataset; because the majority of the student population is Cis-Het, the algorithm defaults to this prediction when symptom profiles overlap. However, this clinical overlap is itself an important finding: while mean group vectors differ significantly (as proven in our MANOVA), individual symptom manifestations are widely shared across identity boundaries.

5.3.4 Visualizing the Discriminant Space

To visualize how these groups separate across the psychological landscape, we mapped the first two discriminant scores and plotted decision boundaries.

Figure 12: LDA Score Plot displaying group separation along LD1 and LD2.

The score plot in Figure 12 visually confirms the heavy overlap in the center of the distribution. However, looking at the 95% confidence ellipses, there is a clear gradient shift: the Cis-Het ellipse is anchored firmly to the left (lower distress along LD1), while the Cis-LGBQ+ and TGNC ellipses drift significantly to the right, confirming higher average symptom burdens.

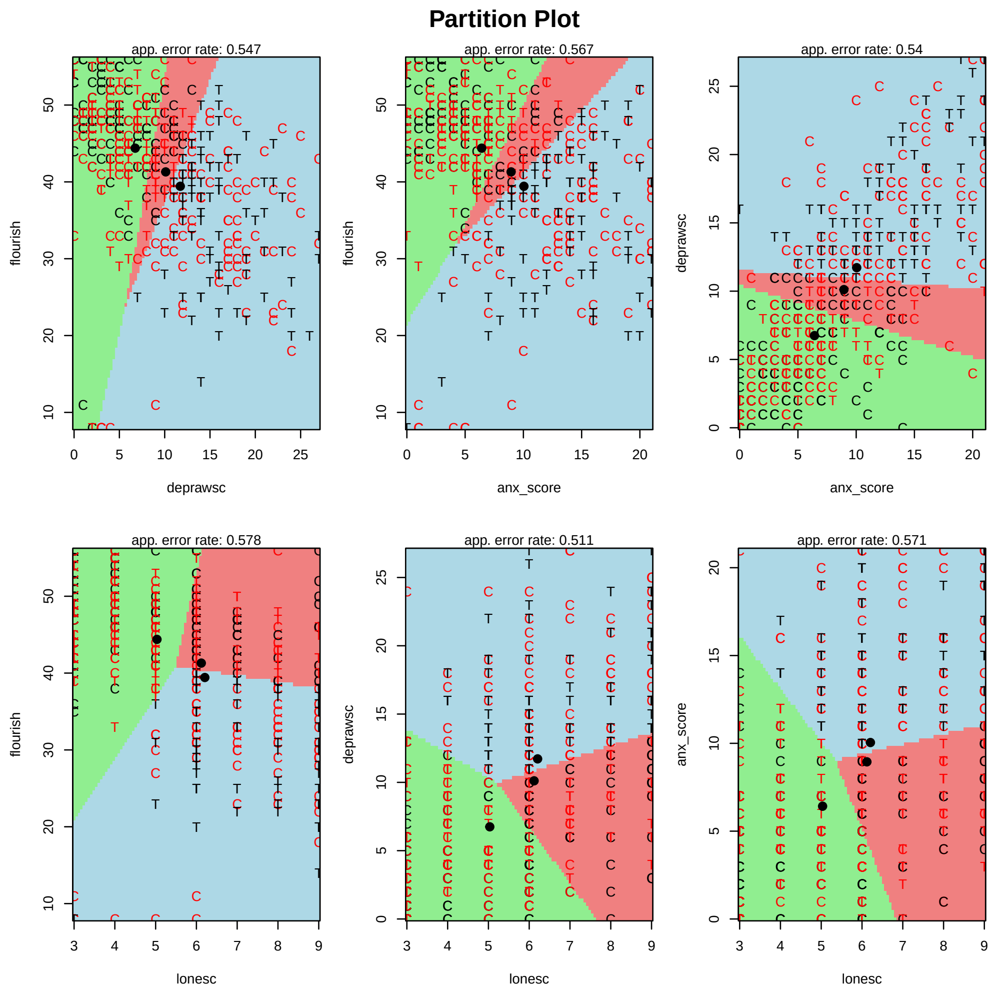

Figure 13: Partition Plot (`partimat`) showing univariate decision boundaries.

Finally, Figure 13 presents a partition plot, visualizing the explicit decision boundary regions for pairwise combinations of our raw variables. These matrices show exactly where the algorithm draws the line between classifying a student as Cis-Het versus marginalized based on specific combinations of symptoms, reinforcing the gradient observed in our structural ordinations.

6 Discussion and Conclusion

This study set out to map the complex, multivariate reality of student mental health using the 2024–2025 Healthy Minds Study. By moving beyond isolated symptom tracking and deploying a sequential multivariate pipeline, we successfully mapped the latent structure of psychological distress, identified the environmental catalysts of severe symptoms, and quantified the disproportionate mental health burden carried by marginalized student populations.

Our dimensionality reduction (PCA) and topological mapping (Ordination) revealed that mental health symptoms among college students do not exist in isolation; they are highly comorbid. Students exist along a continuous gradient of "General Psychological Distress." Crucially, our NMDS environmental overlay demonstrated that while generalized loneliness pulls students toward this dense cluster of comorbid anxiety and depression, acute financial stress (`fincur`) acts as a distinct, powerful vector pointing directly toward severe suicidal ideation and self-harm (`phq9_9`). This underscores that systemic economic burdens require targeted institutional interventions beyond general wellness programs.

Furthermore, our group difference modeling (MANOVA) confirmed a stark reality regarding intersectional identities on campus. Cis-LGBQ+ and TGNC students carry a significantly heavier burden of psychological distress compared to their Cis-Het peers, exhibiting substantially higher depression and lower flourishing scores. While healthy lifestyle covariates, such as increased sleep, provided the largest overall protective effect in our model, they were mathematically insufficient to close the identity-based mental health gap. Similarly, while therapy utilization was a significant predictor of distress, its additive nature in our model suggests it largely serves as a proxy for the population already seeking care, rather than a definitive equalizer of outcomes.

Finally, our predictive modeling (Discriminant Analysis) provided a nuanced clinical takeaway. While the mean mental health vectors of marginalized students are significantly worse than the majority, their individual symptom profiles overlap so heavily with the general student body that a classification algorithm cannot reliably predict sexual or gender identity based on distress alone. This clinical overlap is a vital finding: it proves that while the *distribution* of suffering is heavily skewed against gender and sexual minorities, the *manifestation* of that suffering is universally shared across the student population.

7 Limitations and Further Analysis

While this analysis provides a robust structural overview of student mental health, several avenues remain for future research:

- **Causal Inference and Longitudinal Data:** The HMS dataset used in this analysis is cross-sectional. Consequently, our models are purely associational. For instance, our MANOVA established a strong link between therapy utilization and higher depression, but we cannot isolate the temporal directionality of this relationship. Future research would benefit from applying causal inference frameworks to longitudinal cohorts to evaluate the true average treatment effect (ATE) of specific psychiatric interventions over time.
- **Expanding Intersectionality:** To preserve the statistical power of our models, we aggregated identity into three broad gender and sexual orientation categories. Future analyses should expand this scope by incorporating race, ethnicity, and first-generation college student status, utilizing Multiple Correspondence Analysis (MCA) to map the compounding effects of these intersecting systemic pressures.
- **Granular Institutional Predictors:** Our models utilized broad environmental variables (e.g., hours of sleep, exercise). Further discriminant models could incorporate institutional variables, such as a student's sense of campus belonging, perceived administrative support, or specific types of therapy (e.g., CBT vs. psychiatric medication), to generate actionable, policy-driven insights for university administrations.

A Reproducible R Code

A.1 Data Setup and Cleaning

```
library(tidyverse)
library(haven)
library(corrplot)
library(MASS)
library(car)
library(NbClust)
library(heplots)
library(vegan)
library(PerformanceAnalytics)
library(emmeans)
library(klaR)
library(factoextra)

hms_data <- read_sav("data/hms_data.sav", encoding = 'latin1')
symptoms <- c(paste0('phq9_', 1:9), paste0('gad7_', 1:7))
environment <- c('age', 'lonesc', 'exerc', 'internet_1', 'fincur')

df <- hms_data %>%
  mutate(across(c(gender_trans, gender_queer, gender_nonbin, gender_selfID
     ,
                  sexual_h, sexual_l, sexual_g, sexual_bi, sexual_queer,
                  sexual_quest, sexual_pan, sexual_asexual),
                  ~replace_na(., 0))) %>%
  mutate(
      trans = (gender_trans == 1 | gender_queer == 1 | gender_nonbin == 1
         | gender_selfID == 1),
      queer_flag = (sexual_l == 1 | sexual_g == 1 | sexual_bi == 1 |
         sexual_queer == 1 | sexual_quest == 1 | sexual_pan == 1 | sexual_
         asexual == 1),
      identity_group = case_when(
          trans ~ "TGNC",
          !trans & queer_flag ~ "Cis-LGBQ+",
          !trans & !queer_flag & sexual_h == 1 ~ "Cis-Het",
          TRUE ~ NA_character_
      ),
      identity_group = as.factor(identity_group),
      ther_any = as.factor(ther_any),
      queer = as.factor(queer_flag),
      sleep_wknight = as.numeric(sleep_wknight),
      deprawsc = as.numeric(deprawsc),
      flourish = as.numeric(flourish)
  )
```

A.2 Exploratory Data Analysis and Transformations

```
df %>%
  dplyr::select(flourish, deprawsc, anx_score, lonesc, ed_sde) %>%
  pivot_longer(cols = everything(), names_to = "Variable", values_to = "
     Score") %>%
  ggplot(aes(x = Score)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "black") +
```

```
  facet_wrap(~ Variable, scales = "free") +
  theme_minimal() +
  labs(title = "Histograms of Mental Health Variables", x = "Score", y = "
     Frequency")

pca_data <- df %>%
  dplyr::select(flourish, deprawsc, anx_score, lonesc, ed_sde) %>% drop_na
     ()

corrplot.mixed(cor(pca_data), lower = "number", upper = "ellipse")

set.seed(4747)
pca_plot_sample <- pca_data %>% slice_sample(n = 5000)
chart.Correlation(pca_plot_sample, histogram = TRUE, pch = 19)

cqplot(pca_data, main = "Chi-Square Q-Q Plot for Raw Data")

df_transform <- df %>%
    mutate(
        deprawsc_sqrt = sqrt(deprawsc + 0.5),
        anx_sqrt = sqrt(anx_score + 0.5),
        flourish_sq = flourish^2
    )

pca_data_transform <- df_transform %>%
  dplyr::select(flourish_sq, deprawsc_sqrt, anx_sqrt, lonesc, ed_sde) %>%
  drop_na()

cqplot(pca_data_transform, main = "Chi-Square Q-Q Plot for Transformed
    Data")
```

A.3 Dimensionality Check (PCA)

```
pca_model <- prcomp(pca_data_transform, center = TRUE, scale. = TRUE)
pca_summary <- summary(pca_model)

pca_table <- data.frame(
  Eigenvalue = round(pca_model$sdev^2, 3),
  Variance_Pct = round(pca_summary$importance[2, ] * 100, 1),
  Cumulative_Pct = round(pca_summary$importance[3, ] * 100, 1)
)

fviz_pca_var(pca_model,
             col.var = "contrib",
             gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
             repel = TRUE,
             title = "PCA Biplot: Variable Contributions")
```

A.4 Method 1: Ordination (CA & NMDS)

```
set.seed(4747)
df_subset <- df %>%
  dplyr::select(all_of(symptoms), all_of(environment)) %>%
  drop_na() %>%
```

```
  sample_n(500)

hms_symptoms_subset <- df_subset %>% dplyr::select(all_of(symptoms))
hms_env_subset <- df_subset %>% dplyr::select(all_of(environment))

hms_ca <- cca(hms_symptoms_subset)
plot(hms_ca, choices = c(1, 2), type = "n", main = "CA of Symptoms")
points(hms_ca, display = "sites", col = "lightgrey", pch = 16, cex = 0.5)
text(hms_ca, display = "species", col = "blue", cex = 1, font = 2)

hms_nmds1 <- metaMDS(hms_symptoms_subset, distance = "bray", k = 1, trymax
     = 50)
hms_nmds2 <- metaMDS(hms_symptoms_subset, distance = "bray", k = 2, trymax
     = 50)
hms_nmds3 <- metaMDS(hms_symptoms_subset, distance = "bray", k = 3, trymax
     = 50)

stress_values <- c(hms_nmds1$stress, hms_nmds2$stress, hms_nmds3$stress)
plot(c(1, 2, 3), stress_values, type = "b", pch = 19, col = "blue", lwd =
    2,
     xlab = "Number of Dimensions (k)", ylab = "Stress",
     main = "NMDS Scree Plot for HMS Symptoms",
     ylim = c(0, max(stress_values) + 0.05), xaxt = "n")
axis(1, at = c(1, 2, 3))
abline(h = 0.20, col = "red", lty = 2)

fit_env <- envfit(hms_nmds2, hms_env_subset, permutations = 999)
plot(hms_nmds2, type = "n", main = "NMDS with Environmental Vectors")
points(hms_nmds2, display = "sites", col = "lightgrey", pch = 16, cex =
    0.5)
text(hms_nmds2, display = "species", col = "blue", cex = 0.9, font = 2)
plot(fit_env, col = "red", lwd = 2)

plot(hms_nmds2, type = "n", main = "NMDS Wireplot: Loneliness Gradient")
points(hms_nmds2, display = "sites", col = "lightgrey", pch = 16, cex =
    0.5)
text(hms_nmds2, display = "species", col = "blue", cex = 0.9, font = 2)
surf_lone <- ordisurf(hms_nmds2, hms_env_subset$lonesc, add = TRUE, col =
    "darkgreen", lwd = 2)
```

A.5 Method 2: Two-Way MANOVA

```
df_manova <- df %>%
  dplyr::select(deprawsc, flourish, ther_any, queer, sleep_wknight) %>%
     drop_na()

manova_fit <- lm(cbind(deprawsc, flourish) ~ ther_any * queer, data = df_
    manova)
Manova(manova_fit, type = 'III')
summary.aov(manova_fit)

emm_queer <- emmeans(manova_fit, ~ queer | rep.meas)
emm_ther <- emmeans(manova_fit, ~ ther_any | rep.meas)
summary(pairs(emm_queer), adjust = 'none')
```

```
summary(pairs(emm_ther), adjust = 'none')

model_cov <- lm(cbind(deprawsc, flourish) ~ ther_any + queer + sleep_
    wknight, data = df_manova)
Manova(model_cov, type = "III")
cqplot(model_cov, main = "Chi-Square Q-Q Plot: Multivariate Normality
    Check")

set.seed(4747)
df_subset_mrpp <- df_manova[sample(nrow(df_manova), 2000), ]
dist_matrix <- vegdist(cbind(df_subset_mrpp$deprawsc, df_subset_mrpp$
    flourish), method = 'euclidean')
mrpp(dist_matrix, grouping = df_subset_mrpp$queer)
mrpp(dist_matrix, grouping = df_subset_mrpp$ther_any)
```

A.6 Method 3: Discriminant Analysis

```
df_da <- df %>%
  dplyr::select(identity_group, flourish, deprawsc, anx_score, lonesc, ed_
     sde) %>%
  drop_na() %>%
  filter(identity_group %in% c("Cis-Het", "Cis-LGBQ+", "TGNC")) %>%
  mutate(identity_group = droplevels(identity_group))

boxM(df_da[, -1], df_da$identity_group)

lda_cv <- lda(identity_group ~ ., data = df_da, CV = TRUE)
qda_cv <- qda(identity_group ~ ., data = df_da, CV = TRUE)

step_da <- stepclass(identity_group ~ ., data = as.data.frame(df_da),
    method = "lda", direction = "both")

lda_model <- lda(identity_group ~ ., data = df_da)

sdev <- apply(df_da[, -1], 2, sd)
std_coef <- diag(sdev) %*% lda_model$scaling
rownames(std_coef) <- colnames(df_da)[-1]

conf_matrix <- table(True = df_da$identity_group, Predicted = lda_cv$class
    )
recall <- diag(conf_matrix) / rowSums(conf_matrix)

lda_predict <- predict(lda_model)
df_lda_plot <- data.frame(df_da, lda_predict$x)

ggplot(df_lda_plot, aes(x = LD1, y = LD2, color = identity_group)) +
  geom_point(alpha = 0.3, size = 0.8) +
  stat_ellipse(linewidth = 1.2) +
  scale_color_manual(values = c("Cis-Het" = "#1b9e77", "Cis-LGBQ+" = "#
     d95f02", "TGNC" = "#7570b3")) +
  theme_minimal() +
  labs(title = "LDA Score Plot", x = "Linear Discriminant 1", y = "Linear
     Discriminant 2", color = "Identity Group")
```

```
set.seed(4747)
df_da_sample <- df_da %>% group_by(identity_group) %>% sample_n(150) %>%
   ungroup()
partimat(identity_group ~ flourish + deprawsc + anx_score + lonesc,
         data = df_da_sample, method = "lda",
         image.colors = c("lightgreen", "lightcoral", "lightblue"))
```

B References

Healthy Minds Network (2025). Healthy Minds Study among Colleges and Universities, year (2024–2025) [The Healthy Minds Study]. Healthy Minds Network, University of Michigan, University of California Los Angeles, Boston University, and Wayne State University. https://healthymindsnetwork.org/research/data-for-researchers.

Declaration on the Use of Generative AI: During the preparation of this work, the author used generative AI technologies (specifically, Google Gemini Pro 3.1) to assist with ideation, code development, and manuscript refinement. After using these tools, the authors thoroughly reviewed and edited the results and take full responsibility for the content, code, and conclusions of this report.